

A framework for constraints

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Consider an e.c. enumeration of all prime formulas $\varphi_1, \varphi_2, \dots$

If we define $\mathbb{P}_n((\neg)\varphi_1 \wedge \dots \wedge (\neg)\varphi_k)$ for all such conjunctions C , we'll be able to assign probabilities to all sentences ϕ , by just considering the conjunctions C containing all of ϕ 's prime formulas and defining $\mathbb{P}_n(\phi) := \sum_{C \models \phi} \mathbb{P}_n(C)$. That is, these conjunctions play the role of boolean worlds of arbitrary granularity.

For this, by induction, all we have to do is, given any conjunction $(\neg)\varphi_1 \wedge \dots \wedge (\neg)\varphi_k$ with probability $\mathbb{P}_n((\neg)\varphi_1 \wedge \dots \wedge (\neg)\varphi_k)$, define the conditional

$$\mathbb{P}_n((\neg)\varphi_1 \wedge \dots \wedge (\neg)\varphi_k \wedge \varphi_{k+1}) = p \cdot \mathbb{P}_n((\neg)\varphi_1 \wedge \dots \wedge (\neg)\varphi_k)$$

for some p . And then, of course, we automatically also define

$$\mathbb{P}_n((\neg)\varphi_1 \wedge \dots \wedge (\neg)\varphi_k \wedge \neg\varphi_{k+1}) = (1 - p) \cdot \mathbb{P}_n((\neg)\varphi_1 \wedge \dots \wedge (\neg)\varphi_k)$$

This is nice because many constraints are about conditionals. For example, we'll want

$$\mathbb{P}_n("P_{n+1}(\varphi) = q" \wedge \varphi) := q \cdot \mathbb{P}_n("P_{n+1}(\varphi) = q")$$

So assume for simplicity our base inductor $\mathbb{Q}_n$ has guesses for all conjunctions $(\neg)\varphi_1 \wedge \dots \wedge (\neg)\varphi_n$ (of length n). We want to fix these beliefs, and define by induction all the infinitely many additional values of p above. This, in some sense, naturally captures the small beliefs, and just adds on top of them syntactic properties we might want to enforce, like "in this situation you should believe this other thing", or "this set of probabilities should be max-entropy". Indeed, this immediately yields marginalization.

Coherence

If $\mathbb{Q}_n$ is propositionally coherent and furthermore all our newly defined p are in $[0, 1]$, then $\mathbb{P}_n$ is propositionally coherent.

And actually, a syntactic kind of LUV-coherence follows from propositional coherence, by just considering, for any $[0, 1]$ -LUV X , the prime formulas $X_i :=$ "the i th binary digit of X is 1". Their conjunctions will behave nice, and so also will expectations.

As always with syntactic properties, how much of this knowledge will propagate depends on how much $\mathbb{P}_n$ knows about the relationship between binary digits and other sentences. Maybe we want to enforce properties of the kind

$$\mathbb{P}_n("X > Y" \wedge \neg X_1 \wedge Y_1) = 0$$

and also more complex relationships between different LUVs. But probably this will only make sense if $\mathbb{Q}_n$ already respects these properties in the small beliefs.

Limits

Inspired by the above LUVs, I wanted to say something like [plus other idea about inequality]

$$\mathbb{P}_n(X = Y) = \lim_m \mathbb{P}_n(X_1 \leftrightarrow Y_1 \wedge \dots \wedge X_m \leftrightarrow Y_m)$$

The general version of this ambitious idea is

$$\mathbb{P}_n(\forall m \phi(m)) = \lim_m \mathbb{P}_n(\phi(\underline{1}) \wedge \dots \wedge \phi(\underline{m}))$$

We'd already have the $\leq$ inequality by default, since the sentences in the right follow from the one on the left (given $\mathbb{P}_n$ has some minimally good enforcement or arbitration of implication). But let's see if we can enforce equality.

If we know $\mathbb{P}_n(\forall m \phi(m))$, but not all the $\mathbb{P}_n(\phi(\underline{1}) \wedge \dots \wedge \phi(\underline{m}))$, we can settle on a "max-entropy-ish" distribution for them that respects all constraints we have.¹ For example, if we only have information up to $\phi(\underline{m})$, we can set $r := \mathbb{P}_n(\phi(\underline{1}) \wedge \dots \wedge \phi(\underline{m})) - \mathbb{P}_n(\forall m \phi(m)) \geq 0$, and define

$$\mathbb{P}_n(\phi(\underline{1}) \wedge \dots \wedge \phi(\underline{k})) := \mathbb{P}_n(\forall m \phi(m)) + \frac{r}{2^{k-m}} \quad (1)$$

for $k \geq m$.

We might also have guesses about properties of the whole sequence, not only a finite initial segment. For example, maybe we have a high credence on $\forall m(\phi(m) \vee \phi(m+1))$. I think we'll also be able to find a "good distribution" for the large $\phi(\underline{m})$ satisfying these kinds of constraints, since ultimately they are finitely expressible and we can just tweak the probability decay in (1) in different ways to ensure they are all accommodated. And if they can't all be accommodated, we probably have a contradiction somewhere, and we can enforce syntactically that we don't get that obvious kind of contradiction (that is, the syntactically enforced class of no-contradictions would have to be the same as that of limit constraints).

I think we'll want to enforce these constraints when $\mathbb{Q}_n$ believes in them, not when any extrapolated (and maybe fairly random) part of $\mathbb{P}_n$ believes in them. The latter seems even trickier to construct, we'd need to find the fix-point of "no further corrections". Maybe this is a kind of "detailed reflective knowledge" that only small beliefs, but not large beliefs, should have (since they would add too much information).

Suppose instead that we don't have a guess for $\mathbb{P}_n(\forall m \phi(m))$, and want to find it by looking at the $\mathbb{P}_n(\phi(\underline{1}) \wedge \dots \wedge \phi(\underline{m}))$.

First off, if our guesses about this sequence are incomplete, we can just complete those using available constraints plus max-entropy-ish.

Then, assuming they are complete, their limit will exist because

$$1 \geq \mathbb{P}_n(\phi(\underline{1})) = \mathbb{P}_n(\phi(\underline{0}) \wedge \dots \wedge \phi(\underline{m})) + \sum_{m=1}^1 (\mathbb{P}_n(\phi(\underline{0}) \wedge \dots \wedge \phi(\underline{i})) - \mathbb{P}_n(\phi(\underline{0}) \wedge \dots \wedge \phi(\underline{i+1})))$$

¹Notice that, if $\mathbb{P}_n(\forall m \phi(m)) > 0$, then $\lim_m \mathbb{P}_n(\phi(\underline{1}) \wedge \dots \wedge \neg \phi(\underline{m})) = 0$. That is, it's arbitrarily unlikely for the first counterexample to be arbitrarily large. This is like a "static" version of Gaifman inductivity, and I wonder if other nice properties of LIs have such static counterparts we can enforce.

and so the right sum is always ≤ 1 and the $\mathbb{P}_n(\phi(0) \wedge \dots \wedge \phi(\underline{m}))$ Cauchy converge.

We could run into trouble when the sequence of sentences is self-referential. I haven't thought about this much yet, but I think it collapses down to the usual kind of self-reference. And as above, maybe we don't want to take this into account for purely large sentences?

More generally, what we're trying to do is ensure the induction is well-founded, so that it is not the case, for example, that $\mathbb{P}_n(\forall m \phi(m))$ depends on $\mathbb{P}_n(\phi(1) \wedge \dots \wedge \phi(5))$ and also the other way around. But actually, if we deal with them syntactically (so that we don't care whether $\phi(1)$ is equivalent to talking about the whole sequence or whatever) this won't be a problem: our sentences will also be well-founded in nested quantifiers, of course. Maybe this leads to some unfortunate behavior when our small reasoner actually knows about these equivalences? I'm not sure, but I suspect this will just be the usual "if you know the equivalence you use it, otherwise you don't, but that's the best we can do".

Decision theory

I don't understand what you mean with some of the decision theoretic desiderata. More precisely, I don't understand why we don't obtain them immediately from the other constraints.

For example, consider your example about "Sensible action consequences across branches". Suppose that indeed our small reasoner already knows the structure of this decision problem, that is, $\mathbb{Q}_n("U(obs = \text{coin heads}) = 100\$" | "A(obs = \text{coin tails}) = \text{Pay } 10\$") \approx 1$. Then, what we want is the large beliefs, or whatever procedure we use to condition on action points and look at utility (to pick the best-looking action point), to also reflect this. But what do you mean by this other than $\mathbb{P}_n("U(obs = \text{coin heads}) = 100\$" | "A(obs = \text{coin tails}) = \text{Pay } 10\$") \approx 1$? Which of course we immediately get because "the large beliefs marginalize to the small" (which in this framework we get by definition).

Maybe what you were trying to say is: Our process to decide on action points will probably look very different to just my "extrapolated belief stat" $\mathbb{P}_n$, and we want this wacky thing to still respect such interactions. But then I'm like, don't we want it to respect all small beliefs, and then this is just a particular case?

Alternatively, maybe what you were trying to say is: Even when $\mathbb{Q}_n$ doesn't know about this relationship (the basic shape of the decision problem), we want to hard-code it into the large beliefs / action-choosing procedure. And then I'm like, yes, we can do this syntactically I guess (although if our decision tree is infinite this might present problems, since it amounts to including an infinite amount of information into the belief state), but we probably can just include it in $\mathbb{Q}_n$, and then extrapolate to $\mathbb{P}_n$ as above without caring about these details?

Max entropy

As another example, you mention "you find it difficult to see how max-entropy could be the exact thing we need to make the decision theory work out". And I get the idea: it'd be such a coincidence. But I think maybe any extrapolation into sentences $\mathbb{Q}_n$ doesn't have opinions on (whether it's literally max-entropy, or the off-the-cuff thing I did in equation (1) above, or other versions of that) would work in "extrapolating all of the information $\mathbb{Q}_n$ has to avoid getting exploited, but not

making new bets".

Consider for instance equation (1) above. Maybe Ω knows that this is not how the probabilities actually look like, and because of that we lose some money. But doesn't this fall into the category " Ω quizzed us on some logic questions we still didn't have good guesses on, so too bad we were not logically omniscient"?

Well, I do get how max-ent minimizes expected surprise (thus loss) in normal settings, but I'm not sure there's a well-defined max-ent here to talk about: max-ent in some sentences (like the sequence in (1)) will imply much certainty in other places (like sentences about that sequence).

I think we could only say "try to do max-ent on those sentences that Ω is most likely to quiz us on". And this again feels like "if you have that info you use it, otherwise you're screwed", but there might be something useful here! Since it's not the same for $\mathbb{Q}_n$ to have some encoded knowledge about "what Ω is most likely to quiz" as to actually have a distribution over all those beliefs that is "designed accordingly, to minimize loss". This starts to feel like "fine-tuning the large beliefs to be less exploitable according to the small beliefs", which is very good. And maybe there's also some natural "best guess" given no information, probably simplicity-biased, of the kind "apply max-entropy more to the shorter sentences, and don't care that much about larger sentences being decided for now?".